ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 2 · PRACTICE E

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET second paper (practical, 80 marks): design the breadboard circuit and flowchart of a car-park entrance with a servo gate and a free-bays display, write its Arduino program, then add features to an LDR night-light program.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of THREE questions. Answer ALL the questions.
2. This is a practical paper. Use Fritzing (or another circuit program) for circuits, draw.io (or similar) for flowcharts and the Arduino IDE for programs. If you do not have a computer, draw the circuits and flowcharts neatly by hand and write the programs in your answer book.
3. Save your work regularly. Name every file with the question number and your name or examination number.
4. No circuits need to be built physically; designs are done on the computer or on paper.
5. Programs are not marked strictly on syntax: marks are given for correct logic, correct use of pins and functions, and comments.
6. Arduino Uno pins: digital pins 0–13 (PWM on 3, 5, 6, 9, 10 and 11), analogue inputs A0–A5 (A4 = SDA and A5 = SCL for I2C), 5 V, 3,3 V and GND.

QUESTION 1

> A small office car park has 10 bays. An IR sensor at the entrance detects a car arriving and an IR sensor at the exit detects a car leaving. A servo motor lifts the boom gate at the entrance only if there is a free bay. A 16 × 2 LCD with an I2C module shows the number of free bays. A green LED is on while bays are free, and a red LED comes on when the car park is full.

The following components are available:

- Arduino Uno
- 2 × IR obstacle sensor modules (OUT goes LOW when a car is detected)
- SG90 servo motor (boom gate)
- 16 × 2 LCD with I2C module (VCC, GND, SDA, SCL)
- Green LED, red LED and 2 × 220 Ω resistors
- Breadboard, jumper wires and a 5 V supply

1.1 Using the given components, design the breadboard circuit for the car-park entrance. (14)
HINT: Include both IR sensors on digital inputs; the servo signal on a PWM pin; the LCD's SDA and SCL on A4 and A5; both LEDs with their resistors; and correct 5 V and GND connections for every module.

1.2 Draw a flowchart that shows the logic of the car park. HINT: Include setting up and showing 10 free bays; checking the entrance sensor and opening the gate only if a bay is free; counting cars in and out; updating the LCD; and switching between the green and red LEDs. (14)

1.3 Save and print (or neatly present) the breadboard circuit and the flowchart. (2)

[30]

QUESTION 2

Write the program for the car park in QUESTION 1. Use these connections: entrance IR sensor to pin 2, exit IR sensor to pin 3, servo to pin 9, green LED to pin 6, red LED to pin 7, and the LCD on the I2C pins (address 0x27). Use the LiquidCrystal_I2C and Servo libraries.

2.1 Write an Arduino program that does the following: (18)

• Sets up the pins, the servo (gate closed at 0°) and the LCD, and shows "Free bays: 10".

• When a car arrives at the entrance and a bay is free: opens the gate (90°) for 3 seconds, closes it, and reduces the number of free bays by 1.

• When a car arrives and the car park is full: keeps the gate closed and shows "FULL" on the LCD.

• When the exit sensor detects a car leaving: adds 1 to the free bays (never more than 10).

• Updates the LCD after every change, and switches on the green LED while bays are free and the red LED when the car park is full.

• Waits until a car has moved past a sensor before counting the next one, and is well commented.

2.2 Save and print (or neatly present) the program. (2)

[20]

QUESTION 3

The program below switches on a garden light (an LED on PWM pin 9) when it gets dark. An LDR and a 10 kΩ resistor form a voltage divider on A0; the reading goes DOWN as it gets darker.

```
// Automatic garden light
const int ldrPin = A0;
const int lightPin = 9;
const int darkLevel = 300;      // below this it is dark

void setup() {
  pinMode(lightPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int level = analogRead(ldrPin);
  Serial.println(level);
  if (level < darkLevel) {
    analogWrite(lightPin, 255);   // light fully on
  } else {
    analogWrite(lightPin, 0);     // light off
  }
  delay(500);
}
```

3.1 Change the program so that the dark level is set by a potentiometer on A1 instead of the fixed value 300. The potentiometer reading (0–1023) must be used directly as the dark level. (9)

3.2 Change the program so that, when it gets dark, the light fades in smoothly from off to full brightness over about 2 seconds instead of switching on at once, and fades out when it gets light again. (9)

3.3 Add a push button on pin 2 (using INPUT_PULLUP) that switches a "manual mode" on and off. In manual mode the light stays fully on whatever the light level. (9)

3.4 Save and print (or neatly present) your final program. (3)

[30]

TOTAL: 80

END OF QUESTION PAPER